1. VHDL (10)
    a. Operations

```vhdl
entity one_bit_adder is
    Port ( A : in STD_LOGIC;
           B : in STD_LOGIC;
           Cin : in STD_LOGIC;
           S : out STD_LOGIC;
           Cout : out STD_LOGIC);
end one_bit_adder;

architecture Behavioral of one_bit_adder is

begin

 S <= A xor B xor Cin;
 Cout <= (A and B) or (A and Cin) or (B and Cin);

end Behavioral;
```

```vhdl
entity one_bit_subtractor is
    Port ( A : in STD_LOGIC;
           B : in STD_LOGIC;
           Bin : in STD_LOGIC;
           D : out STD_LOGIC;
           Bout : out STD_LOGIC);
end one_bit_subtractor;

architecture Behavioral of one_bit_subtractor is

begin

 D <= A xor B xor Bin;
 Bout <= (NOT A AND B) or (NOT A AND Bin) or (B AND Bin);

end Behavioral;
```

The comparator and the nand are in the top level design schematic.

    b. Top-Level

```vhdl
--Entity
ENTITY top_level is
  Port (
  sw       : in  STD_LOGIC_VECTOR(15 downto 0);
  clk      : in  STD_LOGIC;
  seg      : out STD_LOGIC_VECTOR(6 downto 0);
  an       : out STD_LOGIC_VECTOR(3 downto 0);
  led0     : out STD_LOGIC;
  led6      : out STD_LOGIC;
  led8      : out STD_LOGIC;
  led10      : out STD_LOGIC

);


end top_level;

ARCHITECTURE Structural of top_level is
-- Components

--One Bit Adder
COMPONENT one_bit_adder IS
    Port ( A : in STD_LOGIC;
           B : in STD_LOGIC;
           Cin : in STD_LOGIC;
           S : out STD_LOGIC;
           Cout : out STD_LOGIC);
end COMPONENT one_bit_adder;

--One Bit Subtractor
COMPONENT one_bit_subtractor IS
    Port ( A : in STD_LOGIC;
           B : in STD_LOGIC;
           Bin : in STD_LOGIC;
           D : out STD_LOGIC;
           Bout : out STD_LOGIC);
end COMPONENT one_bit_subtractor;

--selector
COMPONENT Function_select IS
    PORT ( control : in STD_LOGIC_VECTOR (1 downto 0);
           Input0 : in  STD_LOGIC_VECTOR (3 downto 0);
```

Project Summary | top_level.vhd

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```vhdl
           Input1 : in  STD_LOGIC_VECTOR (3 downto 0);
           Input2 : in  STD_LOGIC_VECTOR (3 downto 0);
           Input3 : in  STD_LOGIC_VECTOR (4 downto 0);
           output : out STD_LOGIC_VECTOR (4 downto 0));
end COMPONENT Function_select;

--Display Driver
COMPONENT Display_driver IS
    Port ( inputs : in  STD_LOGIC_VECTOR (3 downto 0);
          seg_out : out  STD_LOGIC_VECTOR (6 downto 0));
end COMPONENT Display_driver;

--LED Display
COMPONENT LEDdisplay IS
    PORT (
           clk                    : IN  STD_LOGIC;
           seg0,seg1,seg2,seg3    : IN  STD_LOGIC_VECTOR(6 downto 0);
           seg                    : OUT  STD_LOGIC_VECTOR(6  downto 0);
           an                     : OUT STD_LOGIC_VECTOR(3 downto 0));
end COMPONENT LEDdisplay;


--Signal Declaration



--ADDER Signals
SIGNAL as0: STD_LOGIC;
SIGNAL as1: STD_LOGIC;
SIGNAL as2: STD_LOGIC;
SIGNAL as3: STD_LOGIC;

SIGNAL c0: STD_LOGIC;
SIGNAL c1: STD_LOGIC;
SIGNAL c2: STD_LOGIC;
SIGNAL c3: STD_LOGIC;

--DIFF Signals
SIGNAL ds0: STD_LOGIC;
SIGNAL ds1: STD_LOGIC;
SIGNAL ds2: STD_LOGIC;
SIGNAL ds3: STD_LOGIC;
```

Project Summary | top_level.vhd

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```
SIGNAL b0: STD_LOGIC;
SIGNAL b1: STD_LOGIC;
SIGNAL b2: STD_LOGIC;
SIGNAL b3: STD_LOGIC;

--NAND Signals
SIGNAL nand0 : STD_LOGIC;
SIGNAL nand1 : STD_LOGIC;
SIGNAL nand2 : STD_LOGIC;
SIGNAL nand3 : STD_LOGIC;

SIGNAL s0: STD_LOGIC;
SIGNAL s1: STD_LOGIC;
SIGNAL s2: STD_LOGIC;
SIGNAL s3: STD_LOGIC;
SIGNAL s4: STD_LOGIC;

SIGNAL sig_segments1: STD_LOGIC_VECTOR(6 downto 0);
SIGNAL sig_segments2: STD_LOGIC_VECTOR(6 downto 0);



begin

--NAND
nand0 <= not (sw(12) and sw(0));
nand1 <= not (sw(13) and sw(1));
nand2 <= not (sw(14) and sw(2));
nand3 <= not (sw(15) and sw(3));


--Comparator
led10 <=
    (((sw(15) and sw(14) and (not sw(2))) or
      (sw(15) and (not sw(3))) or
      (sw(14) and (not sw(3)) and (not sw(2)))) or
     ((sw(15) xnor sw(3)) and (sw(14) xnor sw(2)) and
      ((sw(13) and sw(12) and (not sw(0))) or
       (sw(13) and (not sw(1))) or
```

Project Summary | top_level.vhd

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```
        (sw(12) and (not sw(1)) and (not sw(0))))))
    when (sw(8) = '1' and sw(9) = '0')
    else '0';

led8 <=
    (sw(15) xnor sw(3)) and (sw(14) xnor sw(2)) and
    (sw(13) xnor sw(1)) and (sw(12) xnor sw(0))
    when (sw(8) = '1' and sw(9) = '0')
    else '0';

led6 <=
    (((not sw(15)) and (not sw(14)) and sw(2)) or
     ((not sw(15)) and sw(3)) or
     ((not sw(14)) and sw(3) and sw(2)) or
     ((sw(15) xnor sw(3)) and (sw(14) xnor sw(2)) and
      (((not sw(13)) and (not sw(12)) and sw(0)) or
       ((not sw(13)) and sw(1)) or
       ((not sw(12)) and sw(1) and sw(0)))))
    when (sw(8) = '1' and sw(9) = '0')
    else '0';

--ADDER
adder0: one_bit_adder
        PORT MAP (
        A => sw(12),
        B => sw(0),
        Cin => sw(6),
        Cout => c0,
        S => as0);

adder1: one_bit_adder
        PORT MAP (
        A => sw(13),
        B => sw(1),
        Cin => c0,
        Cout => c1,
        S => as1);

adder2: one_bit_adder
        PORT MAP (
        A => sw(14),
        B => sw(2),
```

Project Summary | top_level.vhd

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```
        Cin => c1,
        Cout => c2,
        S => as2);

adder3: one_bit_adder
        PORT MAP (
        A => sw(15),
        B => sw(3),
        Cin => c2,
        Cout => c3,
        S => as3);
--ADDER end

--SUBTRACTOR
diff0: one_bit_subtractor
        PORT MAP (
        A => sw(12),
        B => sw(0),
        Bin => '0',
        Bout => b0,
        D => ds0);

diff1: one_bit_subtractor
        PORT MAP (
        A => sw(13),
        B => sw(1),
        Bin => b0,
        Bout => b1,
        D => ds1);

diff2: one_bit_subtractor
        PORT MAP (
        A => sw(14),
        B => sw(2),
        Bin => b1,
        Bout => b2,
        D => ds2);

diff3: one_bit_subtractor
        PORT MAP (
        A => sw(15),
        B => sw(3),
```

Project Summary | top_level.vhd

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```
        Bin => b2,
        Bout => b3,
        D => ds3);

led0 <= b3; -- Underflow light


--SUBTRACTOR END


selector: Function_select
        PORT MAP (
        control(0) => sw(8),
        control(1) => sw(9),

        --NAND
        Input0(0) => nand0,
        Input0(1) => nand1,
        Input0(2) => nand2,
        Input0(3) => nand3,

        --Comparator
        Input1 => "0000",

        --Subtractor
        Input2(0) => ds0,
        Input2(1) => ds1,
        Input2(2) => ds2,
        Input2(3) => ds3,

        --Adder
        Input3(0) => as0,
        Input3(1) => as1,
        Input3(2) => as2,
        Input3(3) => as3,
        Input3(4) => c3,


        output(0) => s0,
        output(1) => s1,
        output(2) => s2,
        output(3) => s3,
```

Project Summary × top_level.vhd ×

D:/Omega_A/Omega_A.srcs/sources_1/new/top_level.vhd

```
        Input3(4) => c3,


        output(0) => s0,
        output(1) => s1,
        output(2) => s2,
        output(3) => s3,
        output(4) => s4);

driver1: Display_driver
        PORT MAP(
        inputs(0) => s0,
        inputs(1) => s1,
        inputs(2) => s2,
        inputs(3) => s3,
        seg_out => sig_segments1);

driver2: Display_driver
        PORT MAP(
        inputs(0)  => s4,
        inputs(1)  => '0',
        inputs(2)  => '0',
        inputs(3)  => '0',
        seg_out => sig_segments2);

LED_display: LEDdisplay
        PORT MAP(
        clk  => clk,
        seg3 => "1111111",        -- blank (unused digit)
        seg2 => "1111111",        -- blank (unused digit)
        seg1 => sig_segments2,    -- left  digit pattern
        seg0 => sig_segments1,    -- right digit pattern
        seg  => seg,              -- top-level output (to board pins)
        an   => an                -- top-level output (to board pins)
        );




end ARCHITECTURE Structural;
```

c. No other functions

d.

The code we copied from Studio 3 and 4 resources were the helper files LEDdisplay.vhd, Display_driver.vhd, and Function_select.vhd.

2. XDC File

Project Summary | constraints_omega_a.xdc

D:/Omega_A/Omega_A.srcs/constrs_1/new/constraints_omega_a.xdc

```xdc
## Clock signal
set_property PACKAGE_PIN W5 [get_ports clk]
    set_property IOSTANDARD LVCMOS33 [get_ports clk]
    create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]

## Switches
set_property PACKAGE_PIN V17 [get_ports {sw[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[0]}]
set_property PACKAGE_PIN V16 [get_ports {sw[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[1]}]
set_property PACKAGE_PIN W16 [get_ports {sw[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[2]}]
set_property PACKAGE_PIN W17 [get_ports {sw[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[3]}]
#set_property PACKAGE_PIN W15 [get_ports {sw[4]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {sw[4]}]
#set_property PACKAGE_PIN V15 [get_ports {sw[5]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {sw[5]}]
set_property PACKAGE_PIN W14 [get_ports {sw[6]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[6]}]
#set_property PACKAGE_PIN W13 [get_ports {sw[7]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {sw[7]}]
set_property PACKAGE_PIN V2 [get_ports {sw[8]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[8]}]
set_property PACKAGE_PIN T3 [get_ports {sw[9]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[9]}]
#set_property PACKAGE_PIN T2 [get_ports {sw[10]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {sw[10]}]
#set_property PACKAGE_PIN R3 [get_ports {sw[11]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {sw[11]}]
set_property PACKAGE_PIN W2 [get_ports {sw[12]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[12]}]
set_property PACKAGE_PIN U1 [get_ports {sw[13]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[13]}]
set_property PACKAGE_PIN T1 [get_ports {sw[14]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[14]}]
set_property PACKAGE_PIN R2 [get_ports {sw[15]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {sw[15]}]


## LEDs
```

Project Summary | constraints_omega_a.xdc

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```xdc
## LEDs
##Carry Out
set_property PACKAGE_PIN U16 [get_ports {led0}]
    set_property IOSTANDARD LVCMOS33 [get_ports {led0}]
#set_property PACKAGE_PIN E19 [get_ports {led[1]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[1]}]
#set_property PACKAGE_PIN U19 [get_ports {led[2]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[2]}]
#set_property PACKAGE_PIN V19 [get_ports {led[3]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[3]}]
#set_property PACKAGE_PIN W18 [get_ports {led[4]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[4]}]
#set_property PACKAGE_PIN U15 [get_ports {led[5]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[5]}]
set_property PACKAGE_PIN U14 [get_ports {led6}]
    set_property IOSTANDARD LVCMOS33 [get_ports {led6}]
#set_property PACKAGE_PIN V14 [get_ports {led[7]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[7]}]
set_property PACKAGE_PIN V13 [get_ports {led8}]
    set_property IOSTANDARD LVCMOS33 [get_ports {led8}]
#set_property PACKAGE_PIN V3 [get_ports {led[9]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[9]}]
set_property PACKAGE_PIN W3 [get_ports {led10}]
    set_property IOSTANDARD LVCMOS33 [get_ports {led10}]
#set_property PACKAGE_PIN U3 [get_ports {led[11]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[11]}]
#set_property PACKAGE_PIN P3 [get_ports {led[12]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[12]}]
#set_property PACKAGE_PIN N3 [get_ports {led[13]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[13]}]
#set_property PACKAGE_PIN P1 [get_ports {led[14]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[14]}]
#set_property PACKAGE_PIN L1 [get_ports {led[15]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[15]}]


##7 segment display
set_property PACKAGE_PIN W7 [get_ports {seg[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[0]}]
set_property PACKAGE_PIN W6 [get_ports {seg[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[1]}]
set_property PACKAGE_PIN U8 [get_ports {seg[2]}]
```

Project Summary | constraints_omega_a.xdc

D:/Omega_A/Omega_A.srcs/constrs_1/new/constraints_omega_a.xdc

```xdc
#set_property PACKAGE_PIN L1 [get_ports {led[15]}]
    #set_property IOSTANDARD LVCMOS33 [get_ports {led[15]}]


##7 segment display
set_property PACKAGE_PIN W7 [get_ports {seg[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[0]}]
set_property PACKAGE_PIN W6 [get_ports {seg[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[1]}]
set_property PACKAGE_PIN U8 [get_ports {seg[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[2]}]
set_property PACKAGE_PIN V8 [get_ports {seg[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[3]}]
set_property PACKAGE_PIN U5 [get_ports {seg[4]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[4]}]
set_property PACKAGE_PIN V5 [get_ports {seg[5]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[5]}]
set_property PACKAGE_PIN U7 [get_ports {seg[6]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {seg[6]}]

#set_property PACKAGE_PIN V7 [get_ports dp]
    #set_property IOSTANDARD LVCMOS33 [get_ports dp]

set_property PACKAGE_PIN U2 [get_ports {an[0]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[0]}]
set_property PACKAGE_PIN U4 [get_ports {an[1]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[1]}]
set_property PACKAGE_PIN V4 [get_ports {an[2]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[2]}]
set_property PACKAGE_PIN W4 [get_ports {an[3]}]
    set_property IOSTANDARD LVCMOS33 [get_ports {an[3]}]
```

3. (4) For each Boolean function implemented, a table of test results should be provided from running the final design on the Basys board. Each table should include:

Adder

2, 4-bit nums	0001 & 0010	1111 & 0000	1010 & 0100	0111 & 0011
Expected (0b)	0011	1111	1110	1010
Expected (0x)	03	0F	0E	0A
Actual (0x)	03	0F	0E	0A

Subtractor

2, 4-bit nums	0001 & 0010	1111 & 0000	1010 & 0100	0111 & 0011
Expected (0b)	0001 (2c)	1111	0110	0100

Expected (0x)	01 (2c)	0F	06	04
Actual (0x)	0F (overflow light on)	0F	06	04

Comparator

2, 4-bit nums	0001 & 0010	1111 & 0000	1010 & 0100	0111 & 0011
Expected	A < B	A > B	A > B	A > B
Actual	A < B	A > B	A > B	A > B

For our comparator, we used LEDs to denote the relation.

Nand

2, 4-bit nums	0001 & 0010	1111 & 0000	1010 & 0100	0111 & 0011
Expected (0b)	1111	1111	1111	1100
Expected (0x)	0F	0F	0F	0C
Actual (0x)	0F	0F	0F	0C

4. Discussion (3)
    a. Jett did the comparator logic and switch select function. Colin did the initial setup, adder, and subtractor logic. Jordan did the nand logic and write-up. We all worked together on troubleshooting and general code structure.
    b. The Behavioral VHDL is the part that processes when the switches are flipped and it controls the output to the LED Display.
    c. The Structural VHDL is the part that connects the components together using Signals.
    d. The aspects we found most challenging were working with the signals, figuring out how to implement, and troubleshooting.
    e. When we ran our final design on the Basys board we had to take a second look at the subtractor. The way our operation works, is a regular subtractor that does not account for overflow. When our leftmost 4 bit input is less than our rightmost 4 bit input, “underflow” occurs. We decided to showcase this by turning on an LED as a warning.